

Testing Framework- Titanic

Capstone++

Objective: To Test the functionality, UI, features, database and model integration of the titanic passenger data survival prediction system.

Scope of testing:

- Testing the UI, validating the forms, templates and urls.
- Testing the integration of models, databases, forms, views and templates.
- Testing the prediction model and accuracy of data.

Test strategy

- Unit Testing
- Model Integration testing
- User Interface testing
- System testing
- Database testing
- User acceptance testing

Testcases

1. Data Preprocessing & Feature Engineering

These tests ensure that the data is cleaned and engineered to train and test the model as expected before it even hits the model.

ID	Feature	Test Scenario	Expected Result	Actual Result
DP-01	Clean data-Missing values	Ensure all the columns have no null, Nan or missing values.	All the empty fields in the data set should be filled using appropriate techniques.	Dataset has no missing values if <code>df.isnull().sum()</code> is run.
DP-02	Handle categorical values	Gender is one hot encoded.	Gender changed to 0 or 1.	All male converted to 0 and female mapped to 1.
DP-03	Feature Extraction	Feature: FamilySize	If SibSp is 1 and Parch is 2, FamilySize should be 4.	Family size is calculated by adding <code>sibsp+parch+1</code> . New column created and data added to dataframe and saved in the file.
DP-04	Feature Extraction	Feature: Age group binning	Assign age group label to the person based on the age. If age(0-2)-Infant, (3-12) child, (13-19) teen, (20-29) young adult, (30-59) adult, 60+ -senior.	Appropriate age group assigned to each person and the dataset is updated accordingly so that it is passed for model training.
DP-05	Model training	Model classifier-choose appropriate model classifier.	Model should split, train and test the data. Compare different models based on accuracy.	Chosen Logistic regression model works without errors and predicts the results.
DP-06	Prediction results	Accuracy	Predicted results has good accuracy score greater than 0.8.	Logistic regression model gives an accuracy score of 0.85.

2. Machine Learning Pipeline

This ensures the model is saved and loaded correctly within the Django environment.

ID	Feature	Test Scenario	Expected Result	Actual Result
ML-01	Model Integration	Pipeline Loading	The Django server starts without errors when loading the .pkl or .joblib model file.	Model pipeline is loaded successfully.
ML-02	Data consistency	Prediction Consistency	The same input data in the Django form yields the same prediction as the Jupyter test.	Form data passed to model and model predicts the survival and probability rates.
ML-03	Prediction results	Probability Output	The model returns a probability (e.g., 0.85) alongside the binary classification (1).	The passenger data , Survival prediction (bool) and probability rate is saved in the database.

3. Django Web Application (UI & Logic)

This covers the user experience and the "Web Logic" vs. "ML Logic" separation.

ID	Feature	Test Scenario	Expected Result	Actual Result
WEB-01	Home page	Home Page Content	Home Page should display Title, project description, Image, how it works and group names as per requirements.	Home page is displayed with project description, Image and how it works and group name.
WEB-02	Prediction form	Form Validation	Submitting empty fields or invalid data types (e.g., text in "Age") triggers a user-friendly error.	Submit empty form or empty fields, user-friendly message is displayed on top of the form.

WEB-03	Result Page	Results Display	After submission, the user is redirected to a results page showing "Survived/Not Survived" and the %, With progress bar and star rating.	Passenger details displayed along with survival and probability. Progress bar appears for probability. User rating selected by user and saved.
WEB-04	Error handling	Error Handling	If the model fails to predict, the user sees a helpful message, not a 500 Server Error/Traceback.	If the model is not loaded, message is displayed to user.

4. Database Persistence & History

This verifies that your data is actually being saved for the long term.

ID	Feature	Test Scenario	Expected Result	Actual result
DB-01	Database persistence	Record Creation	Every prediction creates a new row in the database table.	Prediction results are stored in prediction form in the database.
DB-02	Database persistence	Data Integrity	Database stores: Input features, Result, Probability, and a Timestamp.	Database stores the details entered by the user along with survival, prediction probability and timestamp.
DB-03	History	History Page	Navigating to /history displays a table of all past predictions from the database.	History Page displays the data stored in the prediction record database.

5. Collaboration & Deployment

ID	Test Scenario	Expected Result
GIT-01	Version Control	All team members have contributed commits; the .gitignore correctly excludes the venv and __pycache__.

DOC-01	Documentation	README includes instructions on how to install requirements and run the Django server.
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